



Working with Lists.

1. Iterating over a list using loops.

- Iteration means accessing each element of a list one by one, usually using a loop (like for or while).

- **Using for loop**

A for loop is the most common way to iterate over a list.

Ex:

```
fruits = ['apple', 'banana', 'mango']  
for i in fruits:  
    print(i)
```

```
#output  
apple  
banana  
mango
```

- **Using for loop with range() and len()**

Useful when you need to access both **index** and **value**

Ex:

```
fruits = ['apple', 'banana', 'mango']  
for i in range(len(fruits)):  
    print(i, fruits[i])
```

```
#output  
0 apple  
1 banana  
2 mango
```

➤ **Using while loop**

while loop uses an index to iterate.

2. Sorting and reversing a list using sort(), sorted(), and reverse().

➤ **sort()**

sort() is a list method that sorts the original list in place
(modifies the original list)

Sorts in ascending order by default.

Works only with lists.

Ex:

```
l = [5, 3, 1, 4, 2]
```

```
l.sort()
```

```
print(l)
```

Output:

```
[1, 2, 3, 4, 5]
```

➤ **sorted()**

`sorted()` is a built-in function that returns a new sorted list, without changing the original list.

Can be used on any iterable (lists, tuples)

To sort in descending order: `sorted(list, reverse=True)`

Ex:

```
l = [5, 3, 1, 4, 2]
```

```
l = sorted(l, reverse=True)
```

```
print(l)
```

Output:

```
[5, 4, 3, 2, 1]
```

➤ **reverse()**

`reverse()` is a list method that reverses the order of the elements in place.

It does not sort, just changes the order.

Ex:

```
l= [1, 2, 3, 4, 5]
```

```
l.reverse()
```

```
print(l)
```

Output:

```
[5, 4, 3, 2, 1]
```

3. Basic list manipulations: addition, deletion, updating, and slicing.

➤ **Addition**

- `append()`
adds element at the end
`list.append("hello")`
- `insert()`
adds element at a specific index
`list.insert(1, 'hii')`

- `extend()`
adds elements from another list
`a = [1, 2]`
`b = [3, 4]`
`a.extend(b)`
`print(a)`
output:
`[1, 2, 3, 4]`

➤ Deletion

- `remove()`
removes first matching value
`list.remove('77')`
- `pop()`
removes by index (default is last)
`fruits.pop(0)`

➤ Updating

- (Changing List Elements)

Ex:

```
colors = ['red', 'blue', 'green']  
colors[1] = 'yellow'  
print(colors) # ['red', 'yellow', 'green']
```

➤ Slicing

Accessing a Range of Elements

`list[start:stop]`

ex:

```
numbers = [0, 1, 2, 3, 4, 5]
```

```
print(numbers[1:4])
```

```
# [1, 2, 3]
```

```
print(numbers[:3])
```

```
# [0, 1, 2]
```